

Internship Report (SPE)

2nd year upper cycle (2CS)

Option: Computer Systems (SIQ)

Deep Learning Approaches for Alzheimer's Disease Detection with Focus on Cognitive Impairment Stages

Made by:
Miss. Romaisa Djoghla

Supervised by:
Miss. Nour El Houda Boulkrinat



Table of Contents

List of Acronyms	4
Acknowledgements	5
Introduction	6
1 State of the Art	7
1.1 Introduction	7
1.2 Alzheimer’s Disease	7
1.3 Alzheimer’s Disease Stages	8
1.3.1 Cognitively Normal (CN)	8
1.3.2 Significant Memory Concern (SMC)	8
1.3.3 Mild Cognitive Impairment (MCI)	9
1.3.3.1 Early Mild Cognitive Impairment (EMCI)	9
1.3.3.2 Late Mild Cognitive Impairment (LMCI)	9
1.3.4 Alzheimer’s Disease (AD)	9
1.4 Early Symptoms of Alzheimer’s Disease	9
1.5 Diagnosis of Alzheimer’s Disease in Early Stages	10
1.5.1 Clinical Assessments	10
1.5.2 Biomarkers	10
1.5.3 Neuropsychological Assessments	10
1.5.4 Brain Imaging	10
1.5.4.1 Positron Emission Tomography (PET)	11
1.5.4.2 Magnetic Resonance Imaging (MRI)	11
1.5.4.3 Structural and Functional MRI (sMRI & fMRI)	11
1.6 Early Detection of Alzheimer’s Disease Using Medical Imaging and Machine Learning	12
1.6.1 Decision Trees	12
1.6.2 Support Vector Machines (SVM)	12
1.6.3 Logistic Regression	12
1.6.4 Deep Learning	12
1.7 Detection of Alzheimer’s Disease Using Medical Imaging and Deep Learning . .	13
1.8 Summary of Recent Deep Learning Approaches for Alzheimer’s Disease Detection	14
2 Design and Implementation	15
2.1 Dataset	15
2.2 Data Preprocessing Pipeline	15
2.3 Methodological Overview	17
2.3.1 Weighted Probability-Based Ensemble Method (WPBEM)	17
2.4 Model Design and Training Phases	18
2.4.1 Phase 1: Training Individual Models	18
2.4.2 Phase 2: Comparative Analysis	18

2.4.3	Phase 3: Ensemble Learning (WPBEM)	18
2.5	Architectural Details and Justification	18
2.5.1	ResNet50 (Residual Network)	18
2.5.2	EfficientNetB2	19
2.5.3	DenseNet169	19
2.5.4	DenseNet201	20
2.6	Custom Classification Head	20
2.7	Training Configuration	22
2.8	Experimental Results	22
Conclusion		23
References		25

List of Figures

1.1	a healthy brain VS a brain with advanced Alzheimer’s disease	8
2.1	ResNet50 architecture	19
2.2	DenseNet169 rchitecture	19
2.3	DenseNet201 rchitecture	20

List of Acronyms

AD	Alzheimer's Disease
ADNI	Alzheimer's Disease Neuroimaging Initiative
APP	Amyloid Precursor Protein
Aβ	Amyloid-Beta
BN	Batch Normalization
BOLD	Blood Oxygenation Level-Dependent
CN	Cognitively Normal
CNN	Convolutional Neural Network
CSF	Cerebrospinal Fluid
DL	Deep Learning
DTI	Diffusion Tensor Imaging
EMCI	Early Mild Cognitive Impairment
FA	Fractional Anisotropy
fMRI	Functional Magnetic Resonance Imaging
FPN	Feature Pyramid Network
GAP	Global Average Pooling
LMCI	Late Mild Cognitive Impairment
LASSO	Least Absolute Shrinkage and Selection Operator
MCI	Mild Cognitive Impairment
MD	Mean Diffusivity
ML	Machine Learning
MRI	Magnetic Resonance Imaging
NC	Normal Control
NIfTI	Neuroimaging Informatics Technology Initiative
PET	Positron Emission Tomography
ReLU	Rectified Linear Unit
SMC	Subjective Memory Complaints
sMRI	Structural Magnetic Resonance Imaging
SPE	Stage Pratique en Entreprise
SSIM	Structural Similarity Index Measure
SVM	Support Vector Machine
T1/T2	MRI Weighting Sequences
WPBEM	Weighted Probability-Based Ensemble Method

Acknowledgements

I would like to express my sincere gratitude to all those who supported and guided me throughout this internship and the completion of this report.

First and foremost, I extend my deepest thanks to my supervisor, Miss Nour El Houda Boulkrinat, for her invaluable guidance, patience, and continuous support throughout this project. Her expertise in the field of medical imaging and machine learning provided me with the direction and motivation needed to carry out this work.

I am grateful to the faculty of the École nationale Supérieure d'Informatique (ESI) for the solid academic foundation they have provided throughout my training, which equipped me with the knowledge and skills necessary to undertake this research project.

I would also like to thank the ADNI (Alzheimer's Disease Neuroimaging Initiative) consortium for making their dataset publicly available. This work would not have been possible without this invaluable resource.

Finally, I extend my heartfelt thanks to my family and friends for their unwavering support, encouragement.

Introduction

Alzheimer’s disease (AD) is one of the most devastating neurodegenerative disorders of our time, affecting millions of individuals worldwide and posing a growing challenge to healthcare systems globally. As the population ages, the prevalence of Alzheimer’s disease and related dementias continues to rise, making early and accurate detection increasingly critical. Current treatment options remain limited in their ability to halt disease progression; however, early intervention has been shown to significantly improve quality of life and slow cognitive decline.

The progression of Alzheimer’s disease follows a continuum, beginning with subtle subjective memory complaints and mild cognitive impairment before advancing to full dementia. Accurately identifying and classifying these early stages particularly Early Mild Cognitive Impairment (EMCI) and Late Mild Cognitive Impairment (LMCI) remains one of the most challenging tasks in clinical neuroscience, as the differences between stages can be imperceptible to human observers.

In recent years, deep learning techniques applied to neuroimaging data particularly Magnetic Resonance Imaging (MRI) have emerged as powerful tools for automated disease detection and classification. Convolutional Neural Networks (CNNs) and ensemble methods have demonstrated remarkable performance in distinguishing between Alzheimer’s disease stages, often achieving accuracy levels comparable to clinical specialists.

This internship report, conducted as part of the second-year upper cycle program (2CS) in the Computer Systems (SIQ) option, presents our work on deep learning approaches for Alzheimer’s disease detection with a focus on cognitive impairment stages.

Chapter 1

State of the Art

1.1 Introduction

Alzheimer’s disease (AD) is the most prevalent form of dementia, accounting for nearly 60–80% of cases worldwide. According to the World Health Organization, more than 55 million people are currently living with dementia, a number projected to triple by 2050 due to population aging. AD is not only a medical condition but also a profound societal and economic challenge, as it places an increasing burden on families, caregivers, and healthcare systems.

Early detection of the disease is one of the most pressing challenges in healthcare. Current treatments are limited in their ability to halt progression, but interventions including lifestyle adjustments, symptomatic therapies, and emerging disease-modifying drugs—are significantly more effective when applied in the earliest stages of the condition. For this reason, researchers have focused on understanding and identifying the earliest signs of AD before it reaches full-blown dementia. In recent years, advanced neuroimaging and deep learning approaches have shown promise in detecting subtle brain changes associated with early cognitive impairment. This has positioned machine learning as a vital tool for early classification and prediction of disease progression [Hazarika et al.2023].

1.2 Alzheimer’s Disease

Alzheimer’s disease (AD) is a progressive and irreversible neurodegenerative disorder that causes deterioration of memory, learning, reasoning, and other cognitive abilities, eventually interfering with everyday life and independence. It is the leading cause of dementia worldwide, affecting millions of people and posing a growing healthcare burden. The disease develops slowly, with early symptoms being subtle before progressing to severe impairment, which underlines the importance of early detection and diagnosis [Lanctôt et al.2024].

From a biological perspective, AD is characterized by two major pathological features: the extracellular accumulation of β -amyloid ($A\beta$) plaques and the intracellular formation of tau neurofibrillary tangles. $A\beta$ plaques form when fragments of amyloid precursor protein (APP) are abnormally cleaved and aggregate in the brain. These plaques interfere with neuronal signaling, activate immune cells such as microglia, and initiate inflammatory responses. Tau tangles, on the other hand, occur when tau proteins that normally stabilize microtubules become hyperphosphorylated, causing them to detach and form twisted filaments inside neurons. This disrupts the structural integrity and transport system of neurons, ultimately leading to their death .

These mechanisms cause synaptic dysfunction, inflammation, and neuronal loss, which drive the cognitive decline and behavioral symptoms of AD. Research shows that these changes start

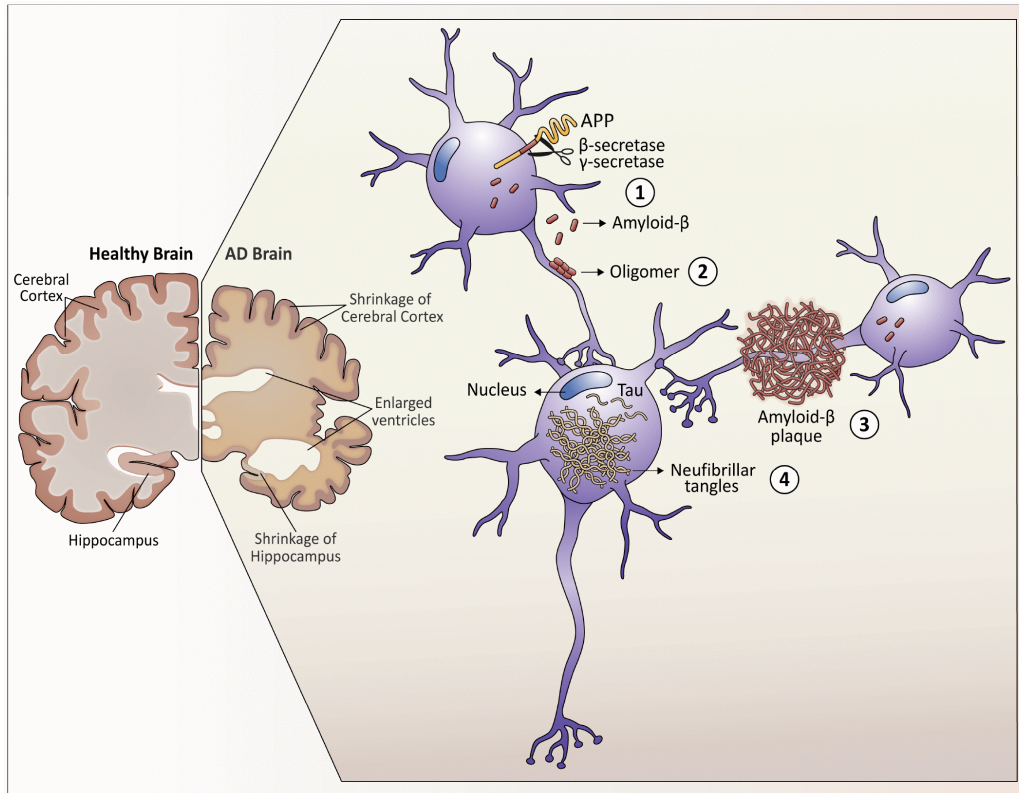


Figure 1.1: a healthy brain VS a brain with advanced Alzheimer's disease

years before clinical signs appear, highlighting the importance of biomarkers for early detection [Yang et al.2023].

1.3 Alzheimer's Disease Stages

The progression of Alzheimer's disease follows a continuum of neurobiological changes that impair cognition and daily independence. Driven by β -amyloid plaques, tau tangles, and related biomarker alterations, these processes start years before symptoms appear. According to [Ramzan et al.2020], the disease can be classified into six stages, each reflecting increasing cognitive and functional decline.

1.3.1 Cognitively Normal (CN)

At this stage, individuals show no measurable cognitive decline. They perform normally in memory and reasoning tests, and daily activities are unaffected. However, subtle neurobiological alterations may already exist, detectable through biomarkers or neuroimaging, years before clinical symptoms appear.

1.3.2 Significant Memory Concern (SMC)

Individuals in this stage report subjective memory problems, such as forgetting words or misplacing objects, even though standardized cognitive assessments remain within normal ranges. SMC is clinically important because it may represent the earliest symptomatic manifestation of Alzheimer's-related pathology.

1.3.3 Mild Cognitive Impairment (MCI)

MCI represents a transitional phase between normal aging and Alzheimer’s dementia. Patients begin to show objective deficits in one or more cognitive domains, most often memory. Although they remain independent in basic daily activities, more complex tasks such as problem-solving or planning may become difficult.

1.3.3.1 Early Mild Cognitive Impairment (EMCI)

In EMCI, cognitive deficits are relatively mild and often limited to episodic memory, such as difficulties recalling recent events or learning new information.

1.3.3.2 Late Mild Cognitive Impairment (LMCI)

LMCI reflects a more advanced level of impairment, extending beyond memory to other cognitive areas, such as attention or executive function. At this point, daily life is increasingly disrupted, and patients often require support with routine or complex activities. LMCI is strongly predictive of conversion to Alzheimer’s dementia.

1.3.4 Alzheimer’s Disease (AD)

This stage is characterized by clinically apparent dementia, with severe impairment in memory, reasoning, orientation, and communication. Behavioral and psychological symptoms, such as agitation or depression, may also emerge. Independence is profoundly compromised, and individuals typically require continuous care.

1.4 Early Symptoms of Alzheimer’s Disease

Alzheimer’s disease (AD) begins with subtle but progressive symptoms that affect cognition and daily life. According to [Kumari, Goel, and Das2022], these early manifestations can be summarized as follows

Memory impairment: One of the earliest symptoms is the inability to retain newly learned information or recall recent conversations and events, often leading to repeated questioning or forgetfulness in daily routines.

Executive function deficits: Difficulties arise in planning, organizing, multitasking, and solving problems, even though patients may still manage most daily activities independently.

Slowed processing speed: Patients may require longer to think, respond, or complete simple tasks, reflecting reduced efficiency in cognitive processing.

Language difficulties: Early AD can cause problems in word-finding and following conversations, which may interfere with effective communication.

Visuospatial challenges: In some cases, individuals experience early difficulties with spatial orientation and recognition of familiar environments or objects.

Reduced learning ability: Even when recall is possible, learning new material becomes slower and requires more effort or repetition.

These early symptoms often progress gradually, and while memory loss is the hallmark, deficits in other domains such as language, executive function, and visuospatial processing can signal the onset of Alzheimer’s disease even before dementia is fully established.

1.5 Diagnosis of Alzheimer’s Disease in Early Stages

Alzheimer’s disease (AD) is a complex neurodegenerative disorder, and its early diagnosis requires a comprehensive and multidimensional evaluation. Recent research highlights that combining multiple assessment strategies can significantly improve the detection of subtle cognitive changes and underlying pathological processes in the initial stages of the disease.

1.5.1 Clinical Assessments

Clinical assessments are the first step in evaluating patients. They involve structured interviews with patients and their families to identify subtle cognitive complaints and behavioral changes. Early-stage symptoms, such as in Subjective Memory Complaints (SMC), are often subtle and may not yet show structural brain changes. Combining clinical evaluations with neuropsychological testing improves early detection [Vanderlip, Stark, and Alzheimer’s Disease Neuroimaging Initiative2024].

1.5.2 Biomarkers

Biomarkers, such as amyloid-beta and tau proteins, can be measured in cerebrospinal fluid (CSF) or blood and provide insights into the early pathological changes of Alzheimer’s disease. These proteins reflect ongoing neurodegenerative processes and can be detected before structural changes appear on MRI, making them particularly useful for identifying early neurodegeneration in individuals with Subjective Memory Complaints (SMC) or Mild Cognitive Impairment (MCI). As highlighted by [Paraskevas and Kapaki2021], monitoring these biomarkers can improve early detection and help track disease progression.

1.5.3 Neuropsychological Assessments

Neuropsychological assessments involve administering detailed batteries of tests to evaluate cognitive functions such as memory, attention, executive function, and language. These tests are particularly valuable in the early stages of Alzheimer’s disease, where subtle cognitive deficits may not yet be visible on brain imaging. Several studies have shown that neuropsychological testing can help detect these early deficits and provide important insights into the likelihood of progression from Subjective Memory Complaints (SMC) to Mild Cognitive Impairment (MCI) or Alzheimer’s disease [Aqeel et al.2022].

1.5.4 Brain Imaging

Brain imaging, including MRI and PET, plays a crucial role in visualizing structural and functional changes associated with Alzheimer’s disease, even at its early stages. MRI allows observation of hippocampal and cortical atrophy, gray matter loss, and cortical thinning, while PET can detect the accumulation of amyloid plaques and tau proteins. Several studies, including [Jiang, Chang, and Zhang2020], [Rehman et al.2024], and [Kang et al.2020], show that combining these modalities improves early detection in patients with Subjective Memory Complaints (SMC) or Early Mild Cognitive Impairment (EMCI).

Brain imaging advances enable the identification of early disease signs, improve understanding of disease progression, and help differentiate Alzheimer’s disease from other neurodegenerative disorders. Early diagnosis often requires a combined approach, integrating imaging, biomarkers, and neuropsychological assessments, particularly in individuals with Subjective Memory Complaints (SMC).

1.5.4.1 Positron Emission Tomography (PET)

PET is a functional imaging technique that visualizes biological abnormalities associated with Alzheimer’s disease. It is particularly useful for detecting amyloid plaques and tau proteins, two major biomarkers of the pathology. These abnormalities can be present in patients with SMC or EMCI before structural changes appear on MRI [Bae et al.2020]. PET provides detailed images that localize affected areas, facilitating early and precise diagnosis.

1.5.4.2 Magnetic Resonance Imaging (MRI)

MRI is widely used for non-invasive brain examination and early detection of structural changes [Bergeron2022]. Advanced MRI analysis highlights:

- **Hippocampal atrophy:** reduced hippocampal volume, a key region for memory, often affected in early stages.
- **Cortical thinning & gray matter loss:** Gray matter segmentation and cortical thickness mapping can detect subtle changes in EMCI and LMCI, enhancing early diagnosis [Aqeel et al.2022]; [Kang et al.2020].
- **Medial temporal lobe and amygdala atrophy:** regions involved in memory and emotions may show volume reduction.
- **Exclusion of other causes:** MRI helps rule out tumors, strokes, or other conditions producing similar symptoms.

MRI can be explored through different planes:

- **Axial:** horizontal view for evaluating regional layout.
- **Sagittal:** side view for hippocampus and internal structures.
- **Coronal:** frontal view for regions involved in memory and emotions.

1.5.4.3 Structural and Functional MRI (sMRI & fMRI)

Advanced MRI approaches combine structural MRI (sMRI) and functional MRI (fMRI) to capture both anatomical and functional brain changes associated with Alzheimer’s disease. This multi-modality imaging provides complementary information beyond standard MRI:

Structural MRI (sMRI): sMRI measures brain anatomy, including gray matter volume, hippocampal atrophy, and cortical thinning. These metrics are particularly useful for detecting subtle changes in individuals with SMC, EMCI, and LMCI [Aqeel et al.2022]; [Kang et al.2020]. sMRI can identify early neurodegeneration that may not yet be visible on conventional MRI scans.

Functional MRI (fMRI): fMRI measures brain activity by detecting changes in blood oxygenation (BOLD signal). It can reveal functional connectivity disruptions in memory and cognitive networks before significant structural changes occur, making it especially valuable in early-stage or prodromal cases [Kang et al.2020].

1.6 Early Detection of Alzheimer’s Disease Using Medical Imaging and Machine Learning

Early detection of Alzheimer’s disease (AD) is crucial to prevent progression from subtle cognitive changes to severe dementia. Machine learning and deep learning techniques, combined with advanced medical imaging, provide effective tools for identifying individuals in the pre-clinical and early stages, including Subjective Memory Complaints (SMC) and Mild Cognitive Impairment (MCI). These approaches can detect subtle structural and functional changes that may be invisible to human observers, improving both diagnosis and prognosis.

1.6.1 Decision Trees

Decision trees are a supervised learning method used to classify data based on features. They recursively split data according to thresholds of relevant variables and can highlight which features are most discriminative. In medical imaging, decision trees can analyze features extracted from MRI, such as regional brain volumes or cortical thickness. For example, [Hamou et al.2011] applied decision trees to identify patterns in MRI features that differentiated cognitively normal individuals from patients with Alzheimer’s disease.

1.6.2 Support Vector Machines (SVM)

SVMs are used to find the optimal hyperplane separating different classes in high-dimensional feature spaces. In MRI studies, features such as hippocampal volume or cortical thickness are used as input to classify patients. [Kumari, Goel, and Das2022] applied SVMs on structural MRI data to distinguish cognitively normal individuals from patients with MCI, demonstrating how nonlinear kernel functions enhance early-stage detection accuracy.

1.6.3 Logistic Regression

Logistic regression models predict probabilities of belonging to a given class based on input features. In neuroimaging, these features might include regional brain volumes or other structural measures. For example, [Rao et al.2011] used logistic regression on MRI-derived features to classify patients with Alzheimer’s disease versus cognitively normal controls, showing the method’s utility for early-stage risk assessment.

1.6.4 Deep Learning

Deep learning methods, especially Convolutional Neural Networks (CNNs), can automatically extract hierarchical features from imaging data:

- **2D and 3D CNNs:** Process MRI slices or volumetric data to capture complex spatial patterns. For instance, [Aqeel et al.2022] used CNNs on gray matter MRI slices to classify cognitively normal versus MCI subjects.
- **Multi-modality MRI (sMRI + fMRI):** Combines structural and functional data, capturing both anatomy and brain activity. [Kang et al.2020] demonstrated improved detection of early cognitive changes using this approach.
- **Transfer Learning:** Pre-trained networks (DenseNet, ResNet) can be fine-tuned on MRI datasets for better performance in early-stage classification **momeni2025transfer**.

- **Graph CNNs:** Analyze cortical thickness networks rather than raw voxels, which can increase sensitivity to subtle structural changes [Jo et al.2020].

1.7 Detection of Alzheimer’s Disease Using Medical Imaging and Deep Learning

In the following, we present recent works applying deep learning techniques for the detection of Alzheimer’s Disease, including the earliest stages SMC (Subjective Memory Complaints) and MCI stages (EMCI, LMCI).

The work conducted by [Şener, Açıcı, and Sümer2025] proposes an approach for early detection of Alzheimer’s disease using EfficientNetB2 combined with Feature Pyramid Network (FPN) and automatic slice selection via SSIM. The model was tested on the ADNI-3 dataset, comprising 627 participants (50 AD, 37 EMCI, 160 MCI, 15 LMCI, 365 CN) with 24,661 T1 MRI images. By automatically selecting the 5 most informative slices with SSIM instead of analyzing all slices, the model achieved remarkable accuracies of 99.45% for AD vs LMCI and 99.19% for AD vs EMCI.

The study by [Naveen and Nagaraj2024] applies transfer learning on pre-trained CNNs (VGG-19, ResNet-50, Inception V3) to MRI images from ADNI for 5-class AD stage classification (NC, EMCI, MCI, LMCI, AD). ResNet-50 delivers the best performance with 98.7% accuracy.

The work conducted by [Fathi et al.2024] proposes an ensemble method for early diagnosis of Alzheimer’s disease using six CNN architectures combined with the WPBEM (Weighted Probability-Based Ensemble Method). The model was tested on the ADNI dataset comprising 721 subjects (132 NC, 157 MCI, 135 EMCI, 131 LMCI, 166 AD). After automatic selection of the 20 most informative slices based on entropy and application of transfer learning, the weighted ensemble achieved remarkable accuracies of 98.57% for NC/AD, 96.37% for NC/EMCI, and 99.83% for LMCI/AD.

The work conducted by [Alorf2025] proposes a hybrid approach for Alzheimer’s disease classification using Transformer and CNN architectures combined with multimodal data fusion. The model was tested on the ADNI dataset comprising 204 subjects (50 CN, 23 SMC, 54 EMCI, 33 LMCI, 6 MCI, 33 AD). Using functional connectivity matrices from rs-fMRI data and clinical information, the hybrid model leverages CNNs for local spatial pattern extraction and Transformers with self-attention for global dependency modeling. The lightweight architecture achieved 96% accuracy for six-stage multiclass classification and 99.65% for CN vs AD binary classification with only 0.879 million parameters.

The study by [Kang et al.2020] proposes a multi-modality deep learning approach for early mild cognitive impairment detection combining structural MRI and DTI data. Tested on 120 ADNI subjects (70 EMCI, 50 NC), the method uses VGG16 CNN with transfer learning for feature extraction, LASSO for feature selection, and SVM for classification. By fusing sMRI, FA, and MD images into RGB channels, the approach captures both macro and microstructural brain changes, achieving 94.2% accuracy for EMCI vs. NC classification.

The study by [Jin et al.2022] proposes a hybrid deep learning method for EMCI vs. LMCI classification using incomplete multimodal data. The three-stage framework combines FF-GAN

with feature fusion-based loss and classification network fine-tuning on 360 ADNI subjects. The approach guides image synthesis toward classification-relevant regions and achieves 85.21% validation accuracy, outperforming traditional methods on incomplete multimodal datasets.

The study by [Wu et al.2025] proposes multiple learning-based approaches for four-stage Alzheimer’s disease progression detection using 3D MRI images. Based on ResNet3D-18 with transfer learning, the framework combines hierarchical, one-versus-one, one-versus-all, and ensemble classification methods on 4,658 ADNI scans. The ensemble approach with nonlinear stacking achieved 95.0% accuracy for CN/EMCI/LMCI/AD classification, demonstrating performance competitive with clinical neurologists and validated through 3D Grad-CAM visualization of disease-relevant brain regions.

The study by [Shamrat et al.2023] introduces AlzheimerNet, a fine-tuned InceptionV3 framework for six-class Alzheimer’s disease stage classification (CN, SMC, EMCI, LMCI, MCI, AD) using 60,000 MRI scans from ADNI. The model integrates global average pooling, dropout, and RMSprop optimization, achieving 98.68% test accuracy and significantly outperforming conventional CNN architectures such as VGG16, ResNet50, and MobileNetV2.

1.8 Summary of Recent Deep Learning Approaches for Alzheimer’s Disease Detection

Table 1.1 presents a comprehensive summary of recent deep learning approaches for Alzheimer’s disease detection, organized by publication year and showing their datasets, image types, models, classification tasks, and achieved accuracy/AUC values.

Year	Reference	Dataset	Image Type	Model	Classification	Accuracy/AUC
2025	[Şener, Açıcı, and Sümer2025]	ADNI-3	MRI 2D T1	EfficientNetB2+FPN with SSIM selection	AD vs LMCI, AD vs EMCI, CN vs AD	99.45%, 99.19%, 98.81%
2025	[Alorf2025]	ADNI (204 subjects)	rs-fMRI + clinical data	Transformer-CNN hybrid with multi-modal fusion	6 stages: CN, SMC, EMCI, MCI, LMCI, AD	Multiclass: 96%, CN vs AD: 99.65%, AD vs EMCI: 96.88%
2025	[Wu et al.2025]	ADNI (4658 scans)	3D MRI (T1-MPRAGE)	Ensemble (ResNet3D-18 + stacking)	4-stages multi-class	95.0%
2024	[Naveen and Nagaraj2024]	ADNI	sMRI	VGG-19, ResNet-50, InceptionV3	CN, EMCI, MCI, LMCI, AD	ResNet-50: 98.7%
2024	[Fathi et al.2024]	ADNI-721 (132 NC, 157 MCI, 135 EMCI, 131 LMCI, 166 AD)	MRI T2 2D (axial slices)	WPBEM (Ensemble of 6 CNNs)	NC/AD, NC/EMCI, EMCI/LMCI, LMCI/AD, 4-way	98.57%, 96.37%, 94.22%, 99.83%
2023	[Shamrat et al.2023]	ADNI (60,000 2D images after augmentation)	MRI T2 2D	AlzheimerNet (InceptionV3 fine-tuned)	CN, SMC, EMCI, MCI, LMCI, AD (6-class)	98.68%
2022	[Jin et al.2022]	ADNI (360 subjects)	MRI + PET (incomplete data)	FF-GAN + ResNet fusion + 3D U-Net	EMCI vs. LMCI	85.21%
2020	[Kang et al.2020]	ADNI (120 subjects)	sMRI + DTI (FA + MD)	CNN (VGG16) + Transfer Learning + SVM	EMCI vs. NC	94.2%

Table 1.1: Summary of Deep Learning Approaches for Alzheimer’s Disease Detection

Chapter 2

Design and Implementation

2.1 Dataset

In this study, I used T1-weighted MRI images from the **ADNI dataset** (Alzheimer’s Disease Neuroimaging Initiative, available at <http://adni.loni.usc.edu>). It provides comprehensive multimodal data to explore the relationships among clinical, cognitive, genetic, and imaging biomarkers of Alzheimer’s disease. For this work, I specifically selected **3T MRI baseline scans**, corresponding to each subject’s first session, to ensure consistent acquisition conditions and avoid longitudinal variability.

The T1-weighted (3T) baseline images were downloaded and preprocessed to construct our working dataset. The dataset used in this work includes four diagnostic classes: **Cognitively Normal (CN)**, **Early Mild Cognitive Impairment (EMCI)**, **Late Mild Cognitive Impairment (LMCI)**, and **Alzheimer’s Disease (AD)**.

Class	Subjects	2D Slices
CN	54	4179
EMCI	60	2800
LMCI	101	3030
AD	69	3008

Table 2.1: Dataset Statistics per Class

2.2 Data Preprocessing Pipeline

The T1-weighted MRI images underwent a comprehensive preprocessing pipeline to ensure standardized, high-quality input for the deep learning models. The preprocessing workflow consisted of twelve sequential stages:

1. **Data Organization:** Raw T1-weighted MRI scans were organized by diagnostic class (CN, EMCI, LMCI, AD) to facilitate class-wise processing and maintain clear dataset structure throughout the pipeline.
2. **DICOM to NIfTI Conversion:** Medical images were initially provided in DICOM format, which stores images with clinical metadata but is not optimal for automated processing. Converting to NIfTI format consolidates multi-slice DICOM series into single 3D volumetric files, simplifying subsequent analysis and ensuring compatibility with neuroimaging processing tools.

3. **Bias Field Correction:** Magnetic field inhomogeneities during MRI acquisition cause uneven intensity distributions across the image, where some regions appear artificially brighter or darker without reflecting true anatomical differences. Bias field correction algorithms compensate for these artifacts by modeling and removing these intensity variations, ensuring that brightness differences reflect actual tissue properties rather than scanner imperfections.
4. **Skull Stripping:** Raw MRI scans contain non-brain tissues such as skull, scalp, eyes, and neck. These structures are irrelevant for Alzheimer’s diagnosis and can introduce noise that confuses the learning process. Skull stripping isolates brain tissue by removing these extraneous structures, allowing the model to focus exclusively on neurologically relevant regions where AD-related changes occur (hippocampus, ventricles, cortex).
5. **Spatial Resampling:** MRI scans acquired from different scanners or protocols have varying spatial resolutions (voxel sizes). For instance, one scan might have $0.8 \times 0.8 \times 0.8$ mm³ voxels while another has $1.2 \times 1.2 \times 1.2$ mm³ voxels. Resampling all volumes to a uniform $1 \times 1 \times 1$ mm³ voxel spacing ensures spatial consistency, enabling fair comparisons across subjects and preventing resolution-dependent biases in model learning.
6. **Cropping:** After skull removal, significant empty space (black background) surrounds the brain. Cropping eliminates this redundant space by tightly framing the brain volume. This reduces computational load during processing and training while ensuring the model focuses on anatomically meaningful regions.
7. **Intensity Normalization:** Different MRI scanners and acquisition parameters produce images with vastly different intensity ranges (e.g., 0–1000 vs. 0–4000). Min-max normalization standardizes all images to the range $[0, 1]$ using the formula: $I_{norm} = \frac{I - I_{min}}{I_{max} - I_{min}}$. This ensures consistent input distributions for the neural network, improving convergence and preventing scanner-dependent biases.
8. **3D to 2D Extraction:** While MRI data are inherently three-dimensional, most pre-trained CNN architectures (ResNet, DenseNet, EfficientNet) are designed for 2D images. Extracting axial (horizontal) slices from each 3D volume converts the data into a format compatible with these architectures. The first and last 10–20 slices, which typically contain minimal brain tissue, are discarded to retain only informative slices.
9. **SSIM-based Slice Selection:** Not all brain slices are equally informative for Alzheimer’s detection. Many consecutive slices are highly similar and redundant. The Structural Similarity Index (SSIM) measures similarity between adjacent slices based on luminance, contrast, and structure. By computing SSIM between consecutive slices, we identify regions with significant structural transitions (e.g., appearance of hippocampus, ventricular changes) and retain 20–40 slices per subject that best capture AD-relevant anatomical variations while discarding redundant information.
10. **Resizing:** Pre-trained CNN models expect fixed input dimensions, typically 224×224 pixels for ImageNet-based architectures. Selected brain slices are resized to this standard dimension using interpolation. Additionally, since these models expect 3-channel (RGB) input but MRI is grayscale, each slice is replicated across three channels to create pseudo-RGB images compatible with transfer learning.
11. **Data Storage:** Preprocessed 2D slices are saved as PNG files in class-specific directories (e.g., CN/, AD/). This organization facilitates efficient data loading during training and enables easy visual inspection of preprocessed data quality.

12. **Data Augmentation:** To improve model generalization and reduce overfitting, training images are augmented with random transformations: rotations ($\pm 10^\circ$), translations ($\pm 10\%$), zoom variations ($\pm 10\%$), and horizontal flips. These augmentations artificially expand the training set by creating realistic variations of existing images, helping the model learn robust features invariant to minor position, orientation, and scale differences.
13. **Dataset Splitting:** To ensure valid performance evaluation, data are split at the *subject level* rather than slice level. This critical distinction prevents data leakage: if multiple slices from the same patient appear in both training and test sets, the model could memorize patient-specific features rather than learning generalizable disease patterns. Subject-level splitting (70% training, 10% validation, 20% test) ensures the model is evaluated on completely unseen patients, providing realistic estimates of clinical performance.

2.3 Methodological Overview

In this project, we adopt an ensemble-based strategy inspired by the paper “*A Deep Learning-Based Ensemble Method for Early Diagnosis of Alzheimer’s Disease using MRI Images*” by Sina Fathi et al. (2023). The authors proposed a Weighted Probability-Based Ensemble Method (WPBEM), which serves as the core approach in this work.

The motivation behind this ensemble technique is to leverage the complementary strengths of multiple deep learning architectures and reduce model bias by weighting each model’s prediction according to its performance.

2.3.1 Weighted Probability-Based Ensemble Method (WPBEM)

WPBEM combines several models based on a weighted sum of their probability outputs, where each model contributes proportionally to its classification accuracy. This ensures that more reliable models have a stronger influence in the final prediction.

Step-by-Step Process

1. **Individual Model Predictions:** Each model produces a probability vector for the target classes (e.g., CN, EMCI, LMCI, AD):

$$P_i = [p_{i,1}, p_{i,2}, \dots, p_{i,K}]$$

where K is the number of classes and i represents the model index.

2. **Assign Weights Based on Accuracy:** Each model is assigned a weight corresponding to its validation accuracy:

$$w_i = \text{accuracy of model } i$$

3. **Weighted Combination of Probabilities:** For each class j :

$$O_j = \sum_i w_i \times p_{i,j}$$

4. **Softmax Normalization:** The final normalized probabilities are obtained via:

$$P_{final,j} = \frac{e^{O_j}}{\sum_k e^{O_k}}$$

5. **Final Prediction:** The predicted class corresponds to:

$$Class_{pred} = \arg \max_j P_{final,j}$$

2.4 Model Design and Training Phases

2.4.1 Phase 1: Training Individual Models

Each CNN architecture (ResNet50, EfficientNetB2, DenseNet169, and DenseNet201) was trained individually using the same dataset and preprocessing pipeline. All models were initialized with ImageNet pre-trained weights (transfer learning).

The training process included:

- Freezing the lower convolutional layers (except the last 30);
- Adding custom fully connected layers for classification;
- Applying class balancing using `compute_class_weight()`;
- Employing data augmentation (rotation, flipping, shifting, zooming);
- Evaluating each model via accuracy, sensitivity, specificity, and confusion matrices.

2.4.2 Phase 2: Comparative Analysis

After training, all models were compared on:

- Binary classification: (CN vs AD, CN vs EMCI, EMCI vs LMCI, LMCI vs AD);
- 3-way classification: (CN / MCI / AD);
- 4-way classification: (CN / EMCI / LMCI / AD).

This phase identifies the strongest architectures and their respective strengths.

2.4.3 Phase 3: Ensemble Learning (WPBEM)

The top-performing models will be combined using the WPBEM method:

$$O_j = \sum_{i=1}^N w_i \times \alpha_{i,j}$$

where w_i is the accuracy weight of model i , and $\alpha_{i,j}$ is the probability of class j predicted by model i . The final prediction corresponds to the class with the highest aggregated probability.

2.5 Architectural Details and Justification

2.5.1 ResNet50 (Residual Network)

Key concept: Skip connections (residual learning). ResNet50 uses residual blocks to enable gradient flow across deep layers.

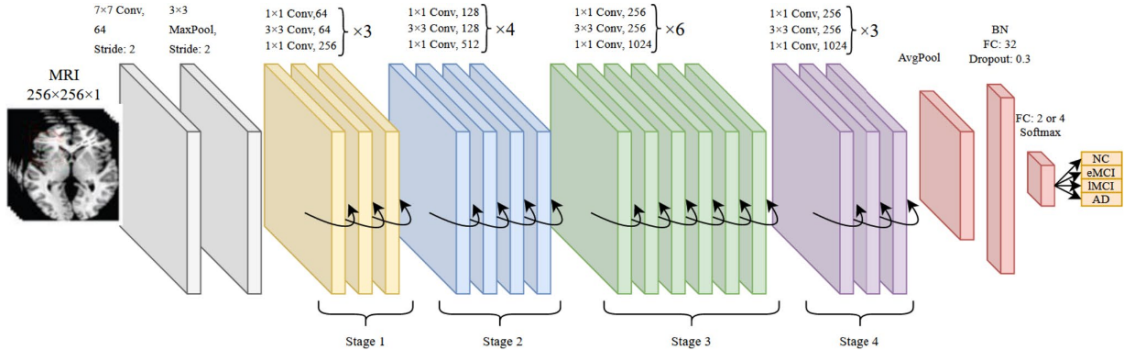


Figure 2.1: ResNet50 architecture

Architecture: Conv7x7 $\rightarrow$ MaxPool $\rightarrow$ Residual Blocks (3+4+6+3) $\rightarrow$ GAP $\rightarrow$ Dense $\rightarrow$ Softmax.

Advantages:

- Prevents vanishing gradients and degradation in deep networks;
- Captures fine structural changes like hippocampal atrophy (important for AD detection).

2.5.2 EfficientNetB2

Key concept: Compound scaling — balancing depth, width, and resolution.

Architecture: Input (260x260x3) $\rightarrow$ MBConv blocks $\rightarrow$ GAP $\rightarrow$ Dense $\rightarrow$ Softmax.

Highlights:

- Uses Squeeze-and-Excitation and Swish activation;
- Excellent trade-off between accuracy and efficiency ($\approx 9M$ parameters);
- Ideal for MRI classification due to compactness and precision.

2.5.3 DenseNet169

Key concept: Dense connectivity each layer receives inputs from all previous layers.

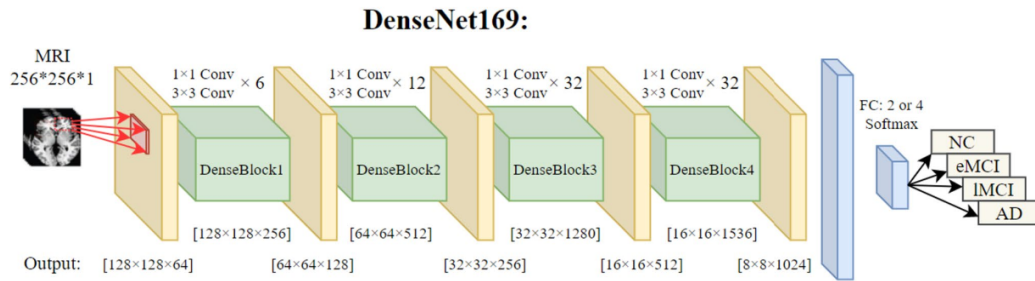


Figure 2.2: DenseNet169 architecture

Architecture: Conv7x7 $\rightarrow$ Dense Blocks (6,12,32,32) $\rightarrow$ GAP $\rightarrow$ Dense $\rightarrow$ Softmax.

Advantages:

- Encourages feature reuse and gradient flow;
- Lightweight yet effective for subtle class distinctions (e.g., EMCI vs LMCI).

2.5.4 DenseNet201

Key concept: Extended dense connectivity with deeper architecture.

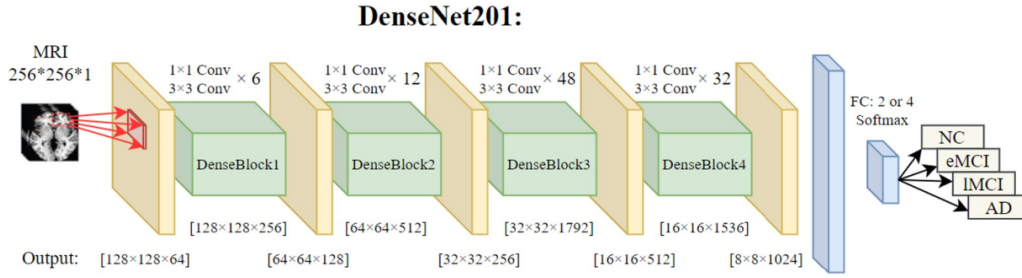


Figure 2.3: DenseNet201 architecture

Architecture: Conv7x7 $\rightarrow$ Dense Blocks (6,12,48,32) $\rightarrow$ GAP $\rightarrow$ Dense $\rightarrow$ Softmax.

Advantages:

- Greater depth enables more sophisticated feature extraction;
- Strong performance on complex multi-class classification tasks.

2.6 Custom Classification Head

A unified classification head was appended to all pre-trained models to adapt the architecture for Alzheimer's disease classification:

Base Model (frozen up to last 30 layers) $\rightarrow$ GAP $\rightarrow$ BN $\rightarrow$ Dense(256, ReLU) $\rightarrow$ Dropout(0.5) $\rightarrow$ BN $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.4) $\rightarrow$ Dense(num_classes, Softmax)

Detailed component description:

- **Base Model (frozen up to last 30 layers):** The weights of the layers pre-trained on ImageNet are preserved and frozen, except for the last 30 layers which remain trainable. This approach allows the model to benefit from generic features learned on ImageNet while enabling adaptation to the specificities of brain MRI images.
- **Global Average Pooling (GAP):** This layer performs spatial averaging across each feature map, reducing spatial dimensions to a one-dimensional vector. Unlike the Flatten layer, GAP significantly reduces the number of parameters and minimizes the risk of overfitting.
- **Batch Normalization (BN):** The first batch normalization layer normalizes activations by centering and reducing their variance. This operation stabilizes training, accelerates convergence, and reduces sensitivity to weight initialization.
- **Dense Layer (256 neurons, ReLU activation):** The first fully connected layer with 256 neurons learns high-level representations from features extracted by the base model. ReLU activation introduces the non-linearity necessary to model complex relationships between features.
- **Dropout Layer (rate of 0.5):** This layer randomly deactivates 50% of neurons during training, forcing the network not to depend on specific neurons. This strong regularization reduces overfitting and improves the model's generalization capability on new images.

- **Batch Normalization (BN):** A second normalization layer is applied to stabilize activations before the second fully connected layer, ensuring uniform gradient distribution during backpropagation.
- **Dense Layer (128 neurons, ReLU activation):** The second fully connected layer progressively reduces dimensionality from 256 to 128 neurons, enabling gradual feature abstraction before final classification. This progressive reduction improves model efficiency.
- **Dropout Layer (rate of 0.4):** An additional dropout with a 40% rate is applied before the output layer to reinforce regularization. This rate, slightly lower than the first dropout, balances regularization and information preservation for final classification.
- **Dense Layer (num_classes neurons, Softmax activation):** The output layer contains a number of neurons corresponding to the number of classes to predict: 2 neurons for binary classification (CN/AD), 3 neurons for three-class classification (CN/MCI/AD), or 4 neurons for four-class classification (CN/EMCI/LMCI/AD). The softmax activation function transforms outputs into normalized probabilities that sum to 1, allowing direct interpretation of the model's confidence for each class.

This architecture balances model capacity with appropriate regularization, enabling effective transfer learning while avoiding overfitting on the relatively limited medical dataset.

Justification:

- Batch Normalization improves stability and convergence;
- Dropout and L2 regularization prevent overfitting;
- Progressive dense layers allow gradual abstraction before classification.

2.7 Training Configuration

Parameter	Value
Optimizer	Adam
Input shape	(224, 224, 3)
Learning rate	0.0001 (initial)
Number of epochs	50 (binary classification)
Batch size	80
Activation functions	ReLU (hidden layers), Softmax (output layer)
Loss function	Categorical Crossentropy
Metrics	Accuracy, Precision, Recall, F1-score
Number of layers (custom head)	5 (2 Dense, 2 BatchNormalization, 2 Dropout + GlobalAveragePooling)
Dropout rates	0.5 and 0.4
Regularization	L2 ($\lambda = 0.01$)
Early stopping	Patience = 20 (monitors validation accuracy)
Image augmentation	Rotation $\pm 8^\circ$, Shift $\pm 8\%$, Zoom $\pm 8\%$, Horizontal flip
Dataset split	70% training, 10% validation, 20% testing

Table 2.2: Training Configuration Parameters

2.8 Experimental Results

[HTML]EFEFEF Stage	ResNet50					DenseNet169					EfficientNetB2					DenseNet201				
	Acc	Pre	Rec	Spe	F1	Acc	Pre	Rec	Spe	F1	Acc	Pre	Rec	Spe	F1	Acc	Pre	Rec	Spe	F1
CN vs. AD	94.8	94.5	95.1	94.4	94.8	96.8	97.0	96.5	96.7	96.8	97.1	96.8	97.4	97.0	97.1	97.4	96.6	98.4	97.3	97.5
CN vs. EMCI	93.9	95.8	91.7	93.3	93.7	95.5	97.7	93.2	95.1	95.4	95.6	97.8	93.5	95.2	95.4	95.9	96.3	94.5	95.8	95.9
EMCI vs. LMCI	84.3	83.0	85.8	83.1	84.4	90.2	89.5	91.0	90.0	90.3	91.3	90.6	92.1	91.0	91.3	91.8	90.9	92.6	91.5	91.8
LMCI vs. AD	98.9	98.7	99.1	99.0	98.9	99.7	99.5	99.9	99.6	99.7	99.8	99.6	99.9	99.7	99.8	99.8	99.7	99.8	99.8	99.8
3-Class (CN, MCI, AD)	79.5	78.9	80.2	79.0	79.5	83.9	83.3	84.4	83.5	83.8	81.3	81.0	81.4	81.1	81.2	84.2	84.0	84.3	84.1	84.1
4-Class (CN, EMCI, LMCI, AD)	77.3	76.8	77.5	77.0	77.2	79.8	79.1	80.0	79.5	79.7	78.2	77.8	78.5	77.9	78.2	81.0	80.7	81.1	80.8	80.9

Table 2.3: Performance comparison across Alzheimer’s classification stages

Conclusion

This internship project investigated deep learning approaches for the automated detection and classification of Alzheimer’s disease stages using T1-weighted MRI images from the ADNI dataset. The work addressed one of the most clinically relevant and technically challenging problems in medical imaging: distinguishing between subtle early-stage cognitive impairment categories, particularly Early Mild Cognitive Impairment (EMCI) and Late Mild Cognitive Impairment (LMCI).

We implemented and evaluated four state-of-the-art CNN architectures ResNet50, EfficientNetB2, DenseNet169, and DenseNet201, under a unified preprocessing pipeline and training configuration. Our preprocessing workflow encompassed twelve sequential stages, from DICOM-to-NIfTI conversion and skull stripping to SSIM-based informative slice selection and data augmentation, ensuring that the models received standardized, high-quality input representative of true brain anatomy.

The experimental results demonstrated that all four architectures achieved strong performance across the classification tasks. For binary classification tasks such as LMCI vs. AD, the best models achieved near-perfect accuracy exceeding 99%, while the more challenging EMCI vs. LMCI discrimination reached accuracies above 91% a clinically significant result given the subtlety of differences between these stages. The four-class classification (CN/EMCI/LMCI/AD) achieved approximately 77–81% accuracy depending on the architecture, consistent with the state of the art on comparable datasets.

Several key observations emerged from this work. First, DenseNet architectures consistently outperformed ResNet50, likely due to their dense connectivity enabling richer feature reuse. Second, SSIM-based slice selection proved highly effective in filtering redundant slices while retaining the most diagnostically informative brain regions. Third, subject-level dataset splitting was essential to avoid data leakage and ensure that reported metrics reflect genuine generalization to unseen patients.

It is important to note that the ensemble learning phase (Phase 3 WPBEM) was designed and formulated as part of the methodological framework but was not fully implemented during this internship due to time constraints. The comparative analysis of individual models (Phase 2) constitutes the primary experimental contribution of this work, and the results already provide a solid foundation for future ensemble integration.

Future work will focus on implementing the Weighted Probability-Based Ensemble Method (WPBEM) by combining the trained models presented here, which is expected to further improve classification performance. Additional directions include exploring three-dimensional volumetric CNN approaches, multi-modal fusion integrating PET and DTI data alongside MRI, and the incorporation of Grad-CAM visualizations to improve model interpretability for clinical use.

Overall, this project demonstrates that deep learning, when combined with rigorous preprocessing and careful architectural selection, offers a powerful and promising framework for early Alzheimer’s disease detection a contribution that may ultimately support earlier clinical intervention and improved patient outcomes.

References

- Alorf, Abdulaziz, (2025). “Transformer and Convolutional Neural Network: A Hybrid Model for Multimodal Data in Multiclass Classification of Alzheimer’s Disease”. In: *Mathematics* 13.10, p. 1548.
- Aqeel, Anza et al., (2022). “A Long Short-Term Memory Biomarker-Based Prediction Framework for Alzheimer’s Disease”. In: *Sensors* 22.4. ISSN: 1424-8220. DOI: [10.3390/s22041475](https://doi.org/10.3390/s22041475). URL: <https://www.mdpi.com/1424-8220/22/4/1475>.
- Bae, Jong Bin et al., (2020). “Identification of Alzheimer’s disease using a convolutional neural network model based on T1-weighted magnetic resonance imaging”. In: *Scientific reports* 10.1, p. 22252.
- Bergeron, David, (2022). “Améliorer le diagnostic des formes atypiques de la maladie d’Alzheimer”. In.
- Fathi, Sina et al., (2024). “A deep learning-based ensemble method for early diagnosis of Alzheimer’s disease using MRI images”. In: *Neuroinformatics* 22.1, pp. 89–105.
- Hamou, Ali et al., (2011). “Cluster analysis of MR imaging in Alzheimer’s disease using decision tree refinement”. In: *International Journal of Artificial Intelligence* 6.S11, pp. 90–99.
- Hazarika, Ruhul Amin et al., (2023). “An Approach for Classification of Alzheimer’s Disease Using Deep Neural Network and Brain Magnetic Resonance Imaging (MRI)”. In: *Electronics* 12.3. ISSN: 2079-9292. DOI: [10.3390/electronics12030676](https://doi.org/10.3390/electronics12030676). URL: <https://www.mdpi.com/2079-9292/12/3/676>.
- Jiang, Xianwei, Liang Chang, and Yu-Dong Zhang, (2020). “Classification of Alzheimer’s disease via eight-layer convolutional neural network with batch normalization and dropout techniques”. In: *Journal of Medical Imaging and Health Informatics* 10.5, pp. 1040–1048.
- Jin, Leiming et al., (2022). “A hybrid deep learning method for early and late mild cognitive impairment diagnosis with incomplete multimodal data”. In: *Frontiers in Neuroinformatics* 16, p. 843566.
- Jo, Taeho et al., (2020). “Deep learning detection of informative features in tau PET for Alzheimer’s disease classification”. In: *BMC bioinformatics* 21.Suppl 21, p. 496.
- Kang, Li et al., (2020). “Identifying early mild cognitive impairment by multi-modality MRI-based deep learning”. In: *Frontiers in aging neuroscience* 12, p. 206.
- Kumari, Rashmi, Shivani Goel, and Subhranil Das, (2022). “Using SVM for Alzheimer’s Disease detection from 3D T1MRI”. In: *2022 IEEE 21st Mediterranean Electrotechnical Conference (MELECON)*. IEEE, pp. 600–604.
- Lanctôt, Krista L. et al., (2024). “Burden of Illness in People with Alzheimer’s Disease: A Systematic Review of Epidemiology, Comorbidities and Mortality”. In: *The Journal of Prevention of Alzheimer’s Disease* 11.1, pp. 97–107. ISSN: 2274-5807. DOI: <https://doi.org/10.14283/jpad.2023.61>. URL: <https://www.sciencedirect.com/science/article/pii/S2274580724001134>.
- Naveen, N and GC Nagaraj, (2024). “Enhancing Early Alzheimer’s Disease Detection: Leveraging Pre-trained Networks and Transfer Learning”. In: *Int. J. Intell. Syst. Appl. (IJISA)* 16.1, pp. 52–69.

- Paraskevas, George P. and Elisabeth Kapaki, (2021). “Cerebrospinal Fluid Biomarkers for Alzheimer’s Disease in the Era of Disease-Modifying Treatments”. In: *Brain Sciences* 11.10. ISSN: 2076-3425. DOI: [10.3390/brainsci11101258](https://doi.org/10.3390/brainsci11101258). URL: <https://www.mdpi.com/2076-3425/11/10/1258>.
- Ramzan, Farheen et al., (2020). “A deep learning approach for automated diagnosis and multi-class classification of Alzheimer’s disease stages using resting-state fMRI and residual neural networks”. In: *Journal of medical systems* 44.2, p. 37.
- Rao, Anil et al., (2011). “Classification of Alzheimer’s Disease from structural MRI using sparse logistic regression with optional spatial regularization”. In: *2011 annual international conference of the IEEE engineering in medicine and biology society*. IEEE, pp. 4499–4502.
- Rehman, Shafiq Ul et al., (2024). “AI-based tool for early detection of Alzheimer’s disease”. In: *Heliyon* 10.8.
- Şener, Begüm, Koray Açıcı, and Emre Sümer, (2025). “Improving early detection of Alzheimer’s disease through MRI slice selection and deep learning techniques”. In: *Scientific Reports* 15.1, p. 29260.
- Shamrat, FM Javed Mehedi et al., (2023). “AlzheimerNet: An effective deep learning based proposition for alzheimer’s disease stages classification from functional brain changes in magnetic resonance images”. In: *IEEE Access* 11, pp. 16376–16395.
- Vanderlip, Casey R., Craig E. L. Stark, and for the Alzheimer’s Disease Neuroimaging Initiative, (2024). “Digital cognitive assessments as low-burden markers for predicting future cognitive decline and tau accumulation across the Alzheimer’s spectrum”. In: *Alzheimer’s & Dementia* 20.10, pp. 6881–6895. DOI: <https://doi.org/10.1002/alz.14154>. eprint: <https://alz-journals.onlinelibrary.wiley.com/doi/pdf/10.1002/alz.14154>. URL: <https://alz-journals.onlinelibrary.wiley.com/doi/abs/10.1002/alz.14154>.
- Wu, Jacky Chung-Hao et al., (2025). “Learning-Based Progression Detection of Alzheimer’s Disease Using 3D MRI Images”. In: *International Journal of Intelligent Systems* 2025.1, p. 3981977.
- Yang, Bowen et al., (2023). “The altered spontaneous neural activity in patients with Parkinson’s disease and its predictive value for the motor improvement of deep brain stimulation”. In: *NeuroImage: Clinical* 38, p. 103430. ISSN: 2213-1582. DOI: <https://doi.org/10.1016/j.nicl.2023.103430>. URL: <https://www.sciencedirect.com/science/article/pii/S2213158223001195>.